

Maximum likelihood estimation of linear models

M514: Econometrics II

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Probability density function

A probability density function (pdf) is a nonnegative function that is used to specify the probability of the random variable falling within a particular range of values. For example, let Z be a random variable whose pdf is given by $f(z)$. Then, the probability that Z takes values between a and b is given by

$$\mathbb{P}\{a \leq Z \leq b\} = \int_a^b f(z)dz$$

In other words, the area under the pdf over a given interval represents the probability that the variable falls within that interval.

Cumulative distribution function

The cumulative distribution function $F(z)$ of a random variable Z is defined as the probability that Z takes on a value less than or equal to z . In other words,

$$F(z) = \mathbb{P}\{-\infty \leq Z \leq z\} = \int_{-\infty}^z f(t)dt$$

Joint pdf and cdf

A joint pdf of two random variables Z_1, Z_2 is similarly defined as a nonnegative function $f(z_1, z_2)$ satisfying

$$\mathbb{P}\{a \leq Z_1 \leq b, c \leq Z_2 \leq d\} = \int_a^b \int_c^d f(z_1, z_2) dz_1 dz_2$$

A joint cdf of Z_1, Z_2 is given by

$$F(z_1, z_2) = \mathbb{P}\{Z_1 \leq z_1, Z_2 \leq z_2\} = \int_{-\infty}^{z_1} \int_{-\infty}^{z_2} f(t_1, t_2) dt_1 dt_2$$

Marginal pdf

Using joint pdf of Z_1, Z_2 , (marginal) pdf of Z_1 can be obtained by

$$f(z_1) = \int_{\mathbb{R}} f(z_1, z_2) dz_2$$

Note that

$$\begin{aligned}\mathbb{P}\{a \leq Z_1 \leq b\} &= \mathbb{P}\{a \leq Z_1 \leq b, -\infty \leq Z_2 \leq \infty\} \\ &= \int_a^b \int_{-\infty}^{\infty} f(z_1, z_2) dz_2 dz_1 = \int_a^b f(z_1) dz_1\end{aligned}$$

as expected.

Independence

The random variables Z_1, Z_2 are said to be independent if the joint pdf separate, i.e.,

$$f(z_1, z_2) = f(z_1)f(z_2)$$

If Z_1, Z_2 are independent, they are uncorrelated. To see this,

$$\begin{aligned}\text{Cov}(Z_1, Z_2) &= \mathbb{E}[(Z_1 - \mathbb{E}[Z_1])(Z_2 - \mathbb{E}[Z_2])] \\ &= \mathbb{E}[Z_1 Z_2] - \mathbb{E}[Z_1]\mathbb{E}[Z_2]\end{aligned}$$

where

$$\begin{aligned}\mathbb{E}[Z_1 Z_2] &= \int_{\mathbb{R}} \int_{\mathbb{R}} z_1 z_2 f(z_1, z_2) dz_1 dz_2 = \int_{\mathbb{R}} \int_{\mathbb{R}} z_1 z_2 f(z_1) f(z_2) dz_1 dz_2 \\ &= \int_{\mathbb{R}} z_1 f(z_1) dz_1 \int_{\mathbb{R}} z_2 f(z_2) dz_2 = \mathbb{E}[Z_1]\mathbb{E}[Z_2]\end{aligned}$$

However, uncorrelatedness does not guarantee independence. Using the law of unconscious statisticians, we may show that independence implies uncorrelatedness of $g(Z_1)$ and $h(Z_2)$ for any g, h .

Likelihood

Let $Z_i, i = 1, \dots, n$ be independent and identically distributed random variables drawn from a common distribution characterized by its pdf $f(z; \theta)$. We assume that the pdf is known up to its parameters θ . Then, the joint pdf of $Z_i, i = 1, \dots, n$ is given by

$$f(z_1, \dots, z_n; \theta) = \prod_{i=1}^n f(z_i; \theta)$$

Suppose we have observations $z_i, i = 1, \dots, n$ from the random variables $Z_i, i = 1, \dots, n$. Then, the likelihood function is given by

$$L(\theta) = f(z_1, \dots, z_n; \theta)$$

Note in the joint pdf of $Z_i, i = 1, \dots, n$, $(z_1, \dots, z_n)$ is used as the arguments (input variables) for the function f , whereas $(z_1, \dots, z_n)$ are the observed samples in the definition of the likelihood.

Maximum likelihood estimator

The maximum likelihood estimator $\hat{\theta}$ is defined as

$$\hat{\theta} = \arg \max_{\theta} L(\theta)$$

The method of maximum likelihood estimates the parameters θ by choosing its values so that the observed data is most probable. Often, it is easier to maximize the log-likelihood $l(\theta) = \log L(\theta)$.

Example 1 (Exponential Distribution)

Suppose the $Z_1, \dots, Z_n$ are iid and follows the exponential distribution $\Gamma(1, \theta)$ with the pdf

$$f(z, \theta) = \theta^{-1} e^{-z/\theta} 1_{\{z > 0\}}$$

The log-likelihood function is given by

$$l(\theta) = \log \prod_{i=1}^n \frac{1}{\theta} e^{-z_i/\theta} = -n \log \theta - \theta^{-1} \sum_{i=1}^n z_i$$

Maximizing $l(\theta)$ w.r.t θ , we get the FOC

$$\frac{\partial l(\theta)}{\partial \theta} = -n\theta^{-1} + \theta^{-2} \sum_{i=1}^n z_i = 0$$

Thus, the MLE is given by $\hat{\theta} = \frac{1}{n} \sum_{i=1}^n z_i$. Since the mean of $\Gamma(1, \theta)$ is given by θ , this is also a sample analogue estimator.

Example 2 (Binomial Distribution)

Let Z be one with probability θ , $0 < \theta < 1$ and zero otherwise. The pmf of Z is

$$p(z; \theta) = \theta^z (1 - \theta)^{1-z}$$

Let $Z_i, i = 1, \dots, n$ be iid samples from Z . Then, the likelihood is given by

$$L(\theta) = \prod_{i=1}^n p(z_i; \theta) = \theta^{\sum_{i=1}^n z_i} (1 - \theta)^{n - \sum_{i=1}^n z_i}$$

and the log-likelihood is

$$l(\theta) = \sum_{i=1}^n z_i \log \theta + \left(n - \sum_{i=1}^n z_i \right) \log(1 - \theta)$$

Example 2 (Binomial Distribution)

The FOC is

$$\frac{\partial l(\theta)}{\partial \theta} = \frac{\sum_{i=1}^n z_i}{\theta} - \frac{n - \sum_{i=1}^n z_i}{1 - \theta} = 0$$

Setting this to 0 and solving for θ , we have

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n z_i$$

Since the mean of a Bernoulli random variable is given by θ , this is also a sample analogue estimator.

Example 3 (Normal Distribution)

Let $Z \sim \mathbb{N}(\mu, \sigma^2)$ with pdf

$$f(z) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(z-\mu)^2}$$

The log-likelihood is given by

$$l(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (z_i - \mu)^2$$

Example 3 (Normal Distribution)

The FOCs are given by

$$\frac{\partial l(\mu, \sigma^2)}{\partial \mu} = -\frac{1}{\sigma^2} \sum_{i=1}^n (z_i - \mu) = 0$$

$$\frac{\partial l(\mu, \sigma^2)}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^n (z_i - \mu)^2 = 0$$

Thus, we obtain

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n z_i$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (z_i - \hat{\mu})^2$$

Example 4 (Uniform Distribution)

Let $Z_i, i = 1, \dots, n$ be iid with the uniform $(0, \theta)$ density, i.e.,

$$f(z) = \frac{1}{\theta} 1_{\{0 < z < \theta\}}$$

Here, $f(z)$ is not differentiable w.r.t. θ . The likelihood function is given by

$$\begin{aligned} L(\theta) &= \theta^{-n} \prod_{i=1}^n 1_{\{0 \leq z_i \leq \theta\}} \\ &= \begin{cases} \theta^{-n} & \text{if } 1_{\{\min_i z_i \geq 0\}} 1_{\{\max_i z_i \leq \theta\}} = 1 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

The function $L(\theta)$ is a decreasing function of θ as long as

$\prod_{i=1}^n 1_{\{0 \leq z_i \leq \theta\}} = 1_{\{\min_i z_i \geq 0\}} 1_{\{\max_i z_i \leq \theta\}} = 1$ and zero otherwise. Thus, $\hat{\theta} = \max_i z_i$.

Note that $\mathbb{E}[Z] = \frac{\theta}{2}$, which gives $\hat{\theta}^{sa} = 2 \frac{1}{n} \sum_{i=1}^n z_i$.

Maximum likelihood estimation of linear regression model

Maximum likelihood estimation of linear regression model

Recall the three classical assumptions on the linear regression model.

1. $(X_i, Y_i), i = 1, \dots, n$ are iid.
2. The variables (X, Y) satisfy the linear regression equation

$$Y = X'\beta + e$$

$$\mathbb{E}[e|X] = 0$$

3.

$$\mathbb{E}[e^2|X] = \sigma^2(X) = \sigma^2$$

is independent of X .

Maximum likelihood estimation of linear regression model

In addition, we assume that

Assumption 4 (Normality)

e is normally distributed.

$$e|X =_d \mathbb{N}(0, \sigma^2)$$

The conditional normality of e implies that Y is also conditionally normally distributed and the distribution is given as

$$Y|X =_d \mathbb{N}(X'\beta, \sigma^2)$$

Maximum likelihood estimation of linear regression model

Let the observations be given by $(X_i, Y_i), i = 1, \dots, n$. By Assumption 1, the observations are jointly normally distributed. Let $\mathbf{Y} = (Y_1, \dots, Y_n)'$, and $\mathbf{X} = (X_1, \dots, X_n)'$. Then, the conditional distribution of $\mathbf{Y}$ given $\mathbf{X}$ is given by

$$\mathbf{Y}|\mathbf{X} =_d \mathbb{N}(\mathbf{X}\beta, \sigma^2 I_n)$$

We want to estimate β and σ^2 using the method of maximum likelihood. The pdf and log-likelihood function of a jointly normal distribution $\mathbb{N}(\mu, \Sigma)$ in $\mathbb{R}^n$ is given by

$$p(x|\theta) = (2\pi)^{-n/2} |\Sigma|^{-1/2} e^{-\frac{1}{2}(x-\mu)'\Sigma^{-1}(x-\mu)}$$
$$l(\theta) = \log p(x|\theta) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \log |\Sigma| - \frac{1}{2} (x - \mu)'\Sigma^{-1}(x - \mu)$$

where $\theta = (\mu, \Sigma)$.

Maximum likelihood estimation of linear regression model

Thus, noting that $\mathbf{Y}|\mathbf{X}$ has mean $\mathbf{X}\beta$ and variance $\sigma^2 I_n$, we get

$$p(\mathbf{Y} = y|\theta) = (2\pi)^{-n/2}(\sigma^2)^{-n/2}e^{-\frac{1}{2\sigma^2}(y-\mathbf{X}\beta)'(y-\mathbf{X}\beta)}$$
$$l(\theta|\mathbf{Y}) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}(\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta)$$

Maximizing $p(\mathbf{Y} = y|\theta)$ w.r.t. β is equivalent to maximizing $l(\theta|\mathbf{Y})$, which is equivalent to minimizing $\frac{1}{2\sigma^2}(\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta)$. This is also equivalent to minimizing $(\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta)$, which is the squared error loss. Thus, $\hat{\beta}^{mle}$ is given by

$$\hat{\beta}^{mle} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

Maximum likelihood estimation of linear regression model

We may also estimate σ^2 by maximizing $l(\theta)$. The FOC is given by

$$-\frac{n}{2} \frac{1}{\sigma^2} + \frac{1}{2(\sigma^2)^2} (\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta) = 0$$

Thus, for any β , $\hat{\sigma}^2$ is given by

$$\hat{\sigma}^2 = \frac{1}{n} (\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta)$$

since we use $\beta = \hat{\beta}$, we have

$$\hat{\sigma}^2 = \frac{1}{n} (\mathbf{Y} - \mathbf{X}\hat{\beta})'(\mathbf{Y} - \mathbf{X}\hat{\beta})$$

Prediction

Suppose we estimate $\hat{\beta}, \hat{\sigma}^2$ are ML estimates from the samples $(X_i, Y_i), i = 1, \dots, n$. We want to predict the value for Y_{n+1} given X_{n+1} . Under the assumptions given in the previous section, Y_{n+1} given X_{n+1} is distributed as

$$Y_{n+1}|X_{n+1} =_d \mathbb{N}(X'_{n+1}\beta, \sigma^2) \quad (1)$$

We predict Y_{n+1} using its conditional mean with the estimated $\hat{\beta}$, i.e.,

$$\begin{aligned}\hat{Y}_{n+1} &= \mathbb{E}[Y_{n+1}|X_{n+1}, \hat{\beta}, \hat{\sigma}^2] \\ &= X'_{n+1}\hat{\beta}\end{aligned}$$

Prediction

Notice that for a standard normal random variable Z , we have

$$\mathbb{P}\{-1.96 \leq Z \leq 1.96\} = 0.95$$

This fact and (1) implies

$$\begin{aligned} & \mathbb{P}\{-1.96 \leq (Y_{n+1} - X'_{n+1}\beta)/\sigma \leq 1.96 | X_{n+1}\} = 0.95 \\ \implies & \mathbb{P}\{X'_{n+1}\beta - 1.96\sigma \leq Y_{n+1} \leq X'_{n+1}\beta + 1.96\sigma | X_{n+1}\} = 0.95 \end{aligned}$$

This gives us not only the point prediction but also the uncertainty associated with the prediction. The 95% prediction interval for Y_{n+1} given X_{n+1} is

$$[X_{n+1}\beta - 1.96\sigma, X_{n+1}\beta + 1.96\sigma]$$

Since we do not know the true values for β and σ , the estimated 95% prediction interval is given by

$$[X_{n+1}\hat{\beta} - 1.96\hat{\sigma}^2, X_{n+1}\hat{\beta} + 1.96\hat{\sigma}^2] \tag{2}$$

Prediction

Under very general assumptions, the MLE estimators are consistent, i.e.,

$$\begin{aligned}\hat{\beta} - \beta &\rightarrow_p 0 \\ \hat{\sigma}^2 - \sigma^2 &\rightarrow_p 0\end{aligned}$$

as $n \rightarrow \infty$. Thus, the prediction interval has the correct coverage asymptotically. In other words, the probability that the true Y_{n+1} is more than $1.96\hat{\sigma}$ below or $1.96\hat{\sigma}$ above our predicted value $\hat{Y}_{n+1}$ is less than 5% asymptotically.

Prediction

However, (2) does not take into account finite sample variance of the estimated $\hat{\beta}$. For this, we need to derive the conditional distribution of $Y_{n+1} - X'_{n+1}\hat{\beta}$. Note that we have

$$\begin{aligned} Y_{n+1} - X'_{n+1}\hat{\beta} &= X'_{n+1}(\beta - \hat{\beta}) + e_{n+1} \\ &= -X'_{n+1}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e} + e_{n+1} \end{aligned}$$

where we've used

$$\begin{aligned} \hat{\beta} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} \\ &= \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e} \end{aligned}$$

Thus, $Y_{n+1} - X'_{n+1}\hat{\beta}$ is jointly normally distributed random variable.

Prediction

The conditional expectation is given by

$$\mathbb{E}[Y_{n+1} - X'_{n+1}\hat{\beta}|\mathbf{X}, X_{n+1}] = 0$$

and the conditional variance is given by

$$\begin{aligned}\text{Var}[Y_{n+1} - X'_{n+1}\hat{\beta}|\mathbf{X}, X_{n+1}] &= \text{Var}(-X'_{n+1}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e} + e_{n+1}|\mathbf{X}, X_{n+1}) \\ &= \text{Var}(X'_{n+1}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e}|\mathbf{X}, X_{n+1}) + \text{Var}(e_{n+1}|\mathbf{X}, X_{n+1}) \\ &\quad + 2\text{Cov}(X'_{n+1}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e}, e_{n+1}|\mathbf{X}, X_{n+1}) \\ &= X'_{n+1}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\text{Var}(\mathbf{e}|\mathbf{X}, X_{n+1})\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}X_{n+1} + \sigma^2 \\ &= \sigma^2 X'_{n+1}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}X_{n+1} + \sigma^2 \\ &= \sigma^2 X'_{n+1}(\mathbf{X}'\mathbf{X})^{-1}X_{n+1} + \sigma^2 \\ &= \sigma^2 (X'_{n+1}(\mathbf{X}'\mathbf{X})^{-1}X_{n+1} + 1)\end{aligned}$$

Prediction

Thus, we have

$$\mathbb{P} \left\{ -1.96 \leq \frac{Y_{n+1} - X'_{n+1} \hat{\beta}}{\sigma \sqrt{X'_{n+1} (\mathbf{X}' \mathbf{X})^{-1} X_{n+1} + 1}} \leq 1.96 \middle| \mathbf{X}, X_{n+1} \right\} = 0.95$$

Inverting this expression, the 95% prediction interval for Y_{n+1} is given by

$$\left[X'_{n+1} \hat{\beta} - 1.96 \sigma \sqrt{X'_{n+1} (\mathbf{X}' \mathbf{X})^{-1} X_{n+1} + 1}, X'_{n+1} \hat{\beta} + 1.96 \sigma \sqrt{X'_{n+1} (\mathbf{X}' \mathbf{X})^{-1} X_{n+1} + 1} \right]$$

Prediction

However, since we do not know σ , we use

$$\left[X'_{n+1} \hat{\beta} - 1.96 \hat{\sigma} \sqrt{X'_{n+1} (\mathbf{X}' \mathbf{X})^{-1} X_{n+1} + 1}, X'_{n+1} \hat{\beta} + 1.96 \hat{\sigma} \sqrt{X'_{n+1} (\mathbf{X}' \mathbf{X})^{-1} X_{n+1} + 1} \right]$$

This is a considerably more precise prediction interval, compared to (2), especially when n is not large. However, we still ignore the uncertainties in $\hat{\sigma}$. Deriving an interval that also takes this into account would be very complicated, without much gains. It is widely believed that estimating β is more difficult than estimating σ .

Remarks

The way that we estimate β is the same in the OLS and MLE framework. In the OLS framework, σ^2 is estimated using the formula

$$\hat{\sigma}^2 = \frac{1}{n-k}(\mathbf{Y} - \mathbf{X}\hat{\beta})'(\mathbf{Y} - \mathbf{X}\hat{\beta})$$

Division by $\frac{1}{n-k}$ is done so that $\hat{\sigma}^2$ is unbiased, but both methods (OLS and MLE) are consistent and are equivalent asymptotically. MLE framework requires a stronger assumption of normality of the error term, but it also gives us richer results in terms of prediction intervals.

Maximum likelihood estimation of linear classification models

Maximum likelihood estimation of linear classification models

We first focus on the case where Y is binary. Let (Y, X) be random with $Y \in \{0, 1\}$ and $X \in \mathbb{R}^k$. As before, define the conditional expectation function error as

$$e := Y - m(X)$$

where $m(x) = \mathbb{E}[Y|X = x]$. Then, we may write

$$Y = m(X) + e$$

$$\mathbb{E}[e|X] = 0$$

There are several characteristics of both $m(x)$ and e that make this model different from the regression framework.

Maximum likelihood estimation of linear classification models

Firstly, note that the conditional expectation function is given by conditional probability.

$$\begin{aligned}m(x) &= \mathbb{E}[Y|X = x] \\&= 1 \cdot \mathbb{P}\{Y = 1|X = x\} + 0 \cdot \mathbb{P}\{Y = 0|X = x\} \\&= \mathbb{P}\{Y = 1|X = x\}\end{aligned}$$

The conditional expectation function always takes values between zero and one. Thus, it is difficult to argue for linear specification of the conditional expectation function.

Maximum likelihood estimation of linear classification models

Secondly, note that the conditional expectation function error is given by

$$e = Y - \mathbb{P}\{Y = 1|X\}$$

The fact that $Y \in \{1, 0\}$ implies that the distribution of e is given by

$$e = \begin{cases} 1 - \mathbb{P}\{Y = 1|X\} & \text{with probability } \mathbb{P}\{Y = 1|X\} \\ -\mathbb{P}\{Y = 1|X\} & \text{with probability } 1 - \mathbb{P}\{Y = 1|X\} \end{cases}$$

The conditional variance is given by

$$\begin{aligned} \text{Var}(e|X) &= \mathbb{E}[e^2|X] \\ &= (1 - \mathbb{P}\{Y = 1|X\})^2 \cdot \mathbb{P}\{Y = 1|X\} + (-\mathbb{P}\{Y = 1|X\})^2 \cdot (1 - \mathbb{P}\{Y = 1|X\}) \\ &= (1 - \mathbb{P}\{Y = 1|X = x\}) \left((1 - \mathbb{P}\{Y = 1|X = x\})\mathbb{P}\{Y = 1|X\} + (\mathbb{P}\{Y = 1|X\})^2 \right) \\ &= (1 - \mathbb{P}\{Y = 1|X\})\mathbb{P}\{Y = 1|X\} \end{aligned}$$

Thus, we cannot assume that e is either a continuous random variable or is conditionally homoskedastic.

Maximum likelihood estimation of linear classification models

Suppose that we ignore these considerations and assume that $m(x)$ is linear in x , i.e., $m(x) = x'\beta$, and that e is normally distributed. As shown in the previous section, maximum likelihood estimates and least squares estimates coincide under these assumptions. Thus, under both the OLS and MLE framework, $\hat{\beta}$ is characterized by

$$\hat{\beta} = \arg \min_{\beta} (\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta)$$

What is wrong with this estimator? Recall that the conditional expectation $m(X) = X'\beta$ has the interpretation of the conditional probability $\mathbb{P}\{Y = 1|X\}$. We want the value $X'\beta$ to reflect how strongly the model believes that $Y = 1$. Suppose $Y = 1$. This means that we want the loss to get smaller as $X'\beta$ gets larger. However, using the squared error loss, the loss actually becomes larger as $X'\beta$ becomes greater when $X'\beta > 1$.

Logit and Probit specification of $m(x)$

Instead of linear specification, we may consider modeling the conditional distribution $m(x) = \mathbb{P}\{Y = 1|X = x\}$ using either standard normal distribution or logistic distribution. This means that we write $m(x) = \Phi(x'\beta)$ where $\Phi(z)$ is standard normal CDF, i.e.,

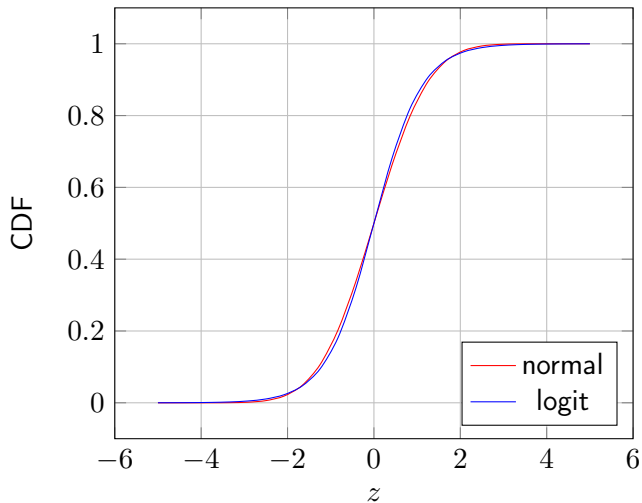
$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-\frac{t^2}{2}} dt$$

or we write $m(x) = \Lambda(x'\beta)$ where

$$\Lambda(z) = \frac{1}{1 + e^{-z}}$$

Here, the functions $\Phi, \Lambda : \mathbb{R} \rightarrow \mathbb{R}$ can be thought of as a nonlinear mapping whose purpose is to map $x'\beta$ to a value in $[0, 1]$.

Logit and Probit specification of $m(x)$



Logit and Probit specification of $m(x)$

Although we use nonlinear mappings, we still call these models 'linear' classification models because $m(x)$ is a function of a single index $x'\beta$, which is linear in x . Linear classification model that uses $m(x) = \Phi(x'\beta)$ is called the probit model, and the model that uses $m(x) = \Lambda(x'\beta)$ is called the logit model (or logistic regression).

Logit and Probit specification of $m(x)$

Under these specifications on $m(x)$, the entire conditional distribution of Y is known up to the parameters β . The conditional pdf of Y is given by

$$\mathbb{P}(Y = y|X) = \begin{cases} \Phi(X'\beta)^y(1 - \Phi(X'\beta))^{1-y} & \text{if probit} \\ \Lambda(X'\beta)^y(1 - \Lambda(X'\beta))^{1-y} & \text{if logit} \end{cases}$$

where $y \in \{0, 1\}$. Suppose we are given i.i.d. samples $(X_i, Y_i), i = 1, \dots, n$. We may construct the joint pdf of (Y_i) as

$$\prod_{i=1}^n p(Y_i = y_i|X_i) = \begin{cases} \prod_{i=1}^n \Phi(X'_i\beta)^{y_i}(1 - \Phi(X'_i\beta))^{1-y_i} & \text{if probit} \\ \prod_{i=1}^n \Lambda(X'_i\beta)^{y_i}(1 - \Lambda(X'_i\beta))^{1-y_i} & \text{if logit} \end{cases}$$

Logit and Probit specification of $m(x)$

and the log-likelihood as

$$\begin{aligned} l(\beta) &= \log \left(\prod_{i=1}^n p(Y_i = y_i | X_i) \right) \\ &= \begin{cases} \sum_{i=1}^n y_i \log \Phi(X_i' \beta) + \sum_{i=1}^n (1 - y_i) (1 - \Phi(X_i' \beta)) & \text{if probit} \\ \sum_{i=1}^n y_i \log \Lambda(X_i' \beta) + \sum_{i=1}^n (1 - y_i) (1 - \Lambda(X_i' \beta)) & \text{if logit} \end{cases} \end{aligned}$$

The maximizer β^* of $l(\beta)$ does not have a closed-form solution. Thus, an iterative numerical optimization method is needed. However, the log-likelihood of both probit and logit model is concave in β and the solution is unique.

Latent variable interpretation

Both probit and logit models have latent variable interpretation. Consider

$$Y^* = X'\beta + u$$
$$Y = \begin{cases} 1 & \text{if } Y^* > 0 \\ 0 & \text{otherwise} \end{cases}$$

Here, the observables are (Y, X) . Y is deterministic given Y^* . The latent variable Y^* is linear in X and has an error term u . The error u has a cumulative distribution function $G(z) = \mathbb{P}\{u \leq z\}$ which is symmetric, i.e.,

$$1 - G(-z) = G(z)$$

Latent variable interpretation

In this model, the conditional probability $\mathbb{P}\{Y = 1|X = x\}$ is given as

$$\begin{aligned}\mathbb{P}\{Y = 1|X = x\} &= \mathbb{P}\{Y^* > 0|X = x\} \\ &= \mathbb{P}\{X'\beta + u > 0|X = x\} \\ &= \mathbb{P}\{u > -X'\beta|X = x\} \\ &= 1 - \mathbb{P}\{u \leq -X'\beta|X = x\} \\ &= \mathbb{P}\{u \leq X'\beta|X = x\} \\ &= G(X'\beta)\end{aligned}$$

where we've used symmetry of $G(z)$ on the fifth equality. Since the conditional probability $\mathbb{P}\{Y = 1|X = x\}$ fully characterizes randomness in Y , the latent variable model with $G(z) = \Phi(z)$ or $G(z) = \Lambda(z)$ is equivalent to the probit or logit model.

Logit model as a minimum loss estimator

Recall that the logit model models the conditional probability $\mathbb{P}\{Y = 1|X = x\}$ as

$$\begin{aligned}\mathbb{P}\{Y = 1|X = x\} &= \Lambda(x'\beta) \\ &= \frac{1}{1 + e^{-x'\beta}}\end{aligned}$$

The size of $x'\beta$ reflects how strongly the model believes that $Y = 1$.

Logit model as a minimum loss estimator

The likelihood function looks like

$$\begin{aligned}l(\beta) &= \log \left(\prod_{i=1}^n p(Y_i = y_i | X_i) \right) \\&= \log \left(\prod_{i=1}^n \left(\frac{1}{1 + e^{-X'_i \beta}} \right)^{y_i} \left(1 - \frac{1}{1 + e^{-X'_i \beta}} \right)^{1-y_i} \right) \\&= \log \left(\prod_{i=1}^n \left(\frac{1}{1 + e^{-X'_i \beta}} \right)^{y_i} \left(\frac{e^{-X'_i \beta}}{1 + e^{-X'_i \beta}} \right)^{1-y_i} \right) \\&= \log \left(\prod_{i=1}^n \left(\frac{1}{1 + e^{-X'_i \beta}} \right)^{y_i} \left(\frac{1}{1 + e^{X'_i \beta}} \right)^{1-y_i} \right)\end{aligned}$$

How does the likelihood behave as we increase $X'_i \beta$? If $Y_i = 1$, the likelihood increases as $X'_i \beta$ increases. If $Y_i = 0$, the likelihood decreases as $X'_i \beta$ increases.

Logit model as a minimum loss estimator

Suppose we still model the conditional probability using logit function, but now code the binary variable Y using $\{-1, 1\}$, instead of $\{0, 1\}$. Then, the log-likelihood becomes

$$\begin{aligned}l(\beta) &= \log \left(\prod_{i=1}^n p(Y_i = y_i | X_i) \right) \\&= \log \left(\prod_{i=1}^n \left(\frac{1}{1 + e^{-y_i X_i' \beta}} \right) \right) \\&= - \sum_{i=1}^n \log \left(1 + e^{-y_i X_i' \beta} \right)\end{aligned}$$

Thus, as we increase $y_i X_i' \beta$, the likelihood also increases.

Logit model as a minimum loss estimator

We define the maximum likelihood estimator $\hat{\beta}$ as

$$\hat{\beta} = \arg \max_{\beta} \left(- \sum_{i=1}^n \log \left(1 + e^{-y_i X_i' \beta} \right) \right)$$

Or, we may regard the negative log-likelihood as our loss function and define a minimum loss estimator as

$$\hat{\beta} = \arg \min_{\beta} \left(\sum_{i=1}^n \log \left(1 + e^{-y_i X_i' \beta} \right) \right)$$

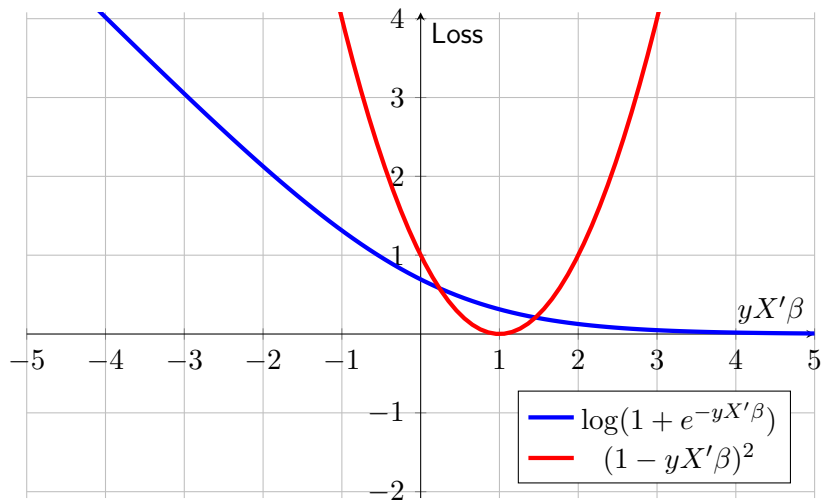
This loss function is called "binomial deviance" loss.

Logit model as a minimum loss estimator

For comparison, suppose $\hat{\beta}$ is estimated by minimizing the squared error loss function.

$$\begin{aligned}\hat{\beta}^{sq} &= \arg \min_{\beta} \left(\sum_{i=1}^n (y_i - X_i' \beta)^2 \right) \\ &= \arg \min_{\beta} \left(\sum_{i=1}^n (1 - y_i X_i' \beta)^2 \right)\end{aligned}$$

Logit model as a minimum loss estimator



Regularized logit model

As with the linear regression model, we may augment the loss function with l_1 penalty term

$$\min_{\beta} \left(\sum_{i=1}^n \log \left(1 + e^{-y_i X_i' \beta} \right) \right) + \lambda \sum_{j=1}^k |\beta_j|$$

or l_2 penalty term

$$\min_{\beta} \left(\sum_{i=1}^n \log \left(1 + e^{-y_i X_i' \beta} \right) \right) + \lambda \beta' \beta$$

Prediction

In the logit model, the conditional probabilities are given by

$$\mathbb{P}\{Y = 1|X = x\} = \frac{1}{1 + e^{-x'\beta}}$$
$$\mathbb{P}\{Y = -1|X = x\} = \frac{e^{-x'\beta}}{1 + e^{-x'\beta}}$$

The linear function $f(x) = x'\beta$ is called the *decision function*. If $f(x) = 0$, we have $\mathbb{P}\{Y = 1|X = x\} = \mathbb{P}\{Y = -1|X = x\} = \frac{1}{2}$. If $f(x) > 0$, we have $\mathbb{P}\{Y = 1|X = x\} > \frac{1}{2}$ and if $f(x) < 0$, we have $\mathbb{P}\{Y = 1|X = x\} < \frac{1}{2}$. Thus, whether to predict a new sample X_{n+1} to be associated with 1 or -1 depends on the sign of $f(X_{n+1})$. The set

$$\{x \in \mathbb{R}^k : f(x) = 0\}$$

is called the *decision boundary*. Notice that $f(x)$ is linear in x , which justifies using the terminology 'linear' classifier.